

1. A 27-year-old man comes to the physician because of a 2-year history of severe muscle cramping and weakness immediately following calisthenics or after jogging a short distance. The cramping and weakness resolve within 24 hours. He also has dark urine during these episodes. Neurologic examination, performed while he is asymptomatic, shows no abnormalities. Which of the following is the most likely diagnosis?
- ☐ A) Alcoholic rhabdomyolysis
 - ☐ B) Hypokalemic periodic paralysis
 - ☐ C) Muscle phosphorylase deficiency (McArdle disease)
 - ☐ D) Myasthenic (Lambert-Eaton) syndrome
 - ☐ E) Polymyositis



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2. A 77-year-old woman with mild dementia, Alzheimer type, is brought to the emergency department by her daughter 3 hours after the sudden onset of fearfulness and combativeness. The patient lives with her daughter and son-in-law. She attempted to smash their television screen with her walker and struck her son-in-law when he tried to stop her. She has hypertension, gastroesophageal reflux disease, and diverticulosis. Her medications are metoprolol, omeprazole, triamterene-hydrochlorothiazide, and donepezil. On arrival, she is hostile and accuses the physician of trying to steal her money. Her temperature is 37.5°C (99.5°F), pulse is 74/min, respirations are 16/min, and blood pressure is 144/82 mm Hg. Physical examination shows restless movements. Mental status examination initially shows hypervigilance; the patient later appears drowsy. She mutters under her breath as if responding to internal stimuli and picks at the air. Her Mini-Mental State Examination score is 10/30. Results of which of the following diagnostic studies are most likely to be abnormal in this patient?
- ☐ A) Liver function tests
 - ☐ B) Measurement of serum vitamin B₁₂ (cobalamin) concentration
 - ☐ C) Serum toxicology screening
 - ☐ D) Thyroid function tests
 - ☐ E) Urinalysis



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3. A 77-year-old woman is brought to the physician by her son because she recently has become unable to take care of herself or walk independently. Her son says that her ability to function has declined during the past 4 months, but she has not had any specific complaints. She has hypertension treated with atenolol and enalapril. Her only other medication is 81-mg aspirin. Six years ago, she had a myocardial infarction. On examination, her speech is confused. Her pulse is 72/min, and blood pressure is 145/86 mm Hg. Examination shows diffuse muscle weakness. Gait testing shows an inability to ambulate. She can recall one of three objects after 2 minutes. Results of laboratory studies are within the reference ranges. An MRI of the brain shows multiple small infarctions throughout both cortices and a few in the thalamus. In addition to adjustments to her antihypertensive medications, administration of which of the following is the most appropriate management?
- ☐ A) Enoxaparin
 - ☐ B) Heparin followed by warfarin
 - ☐ C) Vitamin E
 - ☐ D) Warfarin only
 - ☐ E) No additional medication is indicated



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The response options for the next **2** items are the same. Select one answer for each item in the set.

For each patient with a visual disturbance, select the most likely site of the lesion.

- ☐ A) Cerebellum
- ☐ B) Cranial nerve II
- ☐ C) Cranial nerve III
- ☐ D) Cranial nerve IV
- ☐ E) Cranial nerve VI
- ☐ F) Extraocular muscles
- ☐ G) Medial longitudinal fasciculus
- ☐ H) Neuromuscular junction
- ☐ I) Retina

4. A 62-year-old woman is brought to the emergency department 3 hours after the onset of double vision and left eye pain that she first noted when she awoke. She describes a horizontal separation of images, which is worse for distant as opposed to near vision. She has an 18-year history of type 2 diabetes mellitus. Examination of the left eye shows mild adduction on primary gaze with severe weakness of abduction with conjugate gaze to the left. The remainder of the neurologic examination shows no abnormalities.



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- ☐ F) Extraocular muscles
- ☐ G) Medial longitudinal fasciculus
- ☐ H) Neuromuscular junction
- ☐ I) Retina

5. A 35-year-old man comes to the physician 1 day after the sudden onset of double vision. He describes seeing two objects side by side that are farthest apart when he looks to the right. Examination shows the eyes move normally when looking to the left. When looking to the right, the left eye does not fully adduct, and there is nystagmus in the abducting eye. The remainder of the neurologic examination shows no abnormalities.



6. A 47-year-old woman with chronic alcoholism is brought to the emergency department acutely intoxicated. Other than her intoxication, physical and neurologic examinations show no abnormalities. Which of the following complications is most likely to be prevented by immediate administration of vitamin B₁ (thiamine)?

- ☐ A) Acute alcoholic myopathy
- ☐ B) Anterograde amnesia
- ☐ C) Burning paresthesias of the feet
- ☐ D) Hallucinations
- ☐ E) Seizures



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7. An 82-year-old woman has had a chronic bitemporal headache and diffuse achiness of the shoulders for 6 months. She has no history of serious illness and takes no medications. Her temperature is 37°C (98.6°F), pulse is 80/min, respirations are 12/min, and blood pressure is 150/85 mm Hg. There is mild deformity of the distal interphalangeal joints; range of motion is full, and there is no edema or erythema of the joints. Examination shows normal fundi. Neurologic examination shows no focal findings. Erythrocyte sedimentation rate is 120 mm/h. Which of the following is the most likely diagnosis?
- ☐ A) Ankylosing spondylitis
 - ☐ B) Fibromyositis
 - ☐ C) Polymyalgia rheumatica
 - ☐ D) Polymyositis
 - ☐ E) Rheumatoid arthritis



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8. A 57-year-old man comes to the physician because of a 1-year history of progressive difficulty hearing, especially on the right. He frequently needs to ask people to repeat what they are saying, especially children. His partner has told him that he has been increasing the volume of their television at home. The patient does not recall any related trauma or previous problems with his ears. For 20 years, he has managed a nightclub that features live performances. He has no history of serious illness. He occasionally takes aspirin, ibuprofen, or acetaminophen for headache or low back pain. There is no family history of hearing loss. His pulse is 76/min and regular, and blood pressure is 124/74 mm Hg. On examination, the external ear canals and tympanic membranes appear normal. There is a small amount of cerumen in both ears. He has difficulty hearing whispered voice bilaterally, especially on the right. A vibrating tuning fork placed on the forehead is heard better on the left. Air conduction is greater than bone conduction bilaterally but clearer on the left. Which of the following is the most likely explanation for these findings?

- ☐ A) Adverse effect of aspirin
- ☐ B) Loss of mobility of the auditory ossicles
- ☐ C) Meniere disease
- ☐ D) Scarring of the tympanic membranes
- ☐ E) Sensorineural hearing loss



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9. A 77-year-old man comes to the physician because of a 3-week history of low-grade fever, fatigue, severe headache, sensitivity to bright light, and pain in the hips and shoulders. He has also had a 3.2-kg (7-lb) weight loss due to a decreased appetite during this period. He appears ill. His temperature is 38.2°C (100.8°F). Examination shows mild photophobia but no meningeal signs. There is moderate pain on palpation of the large muscle groups of the shoulders and hips. The remainder of the examination shows no abnormalities. His hemoglobin concentration is 11.1 g/dL, leukocyte count is 11,600/mm³ with a normal differential, and erythrocyte sedimentation rate is 85 mm/h. Which of the following is the most appropriate next step in diagnosis?

- ☐ A) Cerebral angiography
- ☐ B) MRI of the brain
- ☐ C) Lumbar puncture
- ☐ D) Nerve and muscle biopsy
- ☐ E) Temporal artery biopsy



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10. A previously healthy 42-year-old man comes to the physician because of a 1-week history of moderate back pain. The pain originates in his right buttock and radiates down the posterior thigh and calf to the back of his foot. He has not had any changes in bladder or bowel function. He has no history of trauma to the back. His pulse is 82/min, respirations are 16/min, and blood pressure is 132/84 mm Hg. Sensation to light touch is decreased over the right posterior leg and lateral foot. Muscle strength and deep tendon reflexes are normal. Which of the following is the most appropriate next step in management?
- ☐ A) Myelography
 - ☐ B) Gabapentin therapy
 - ☐ C) Ibuprofen therapy
 - ☐ D) Prednisone therapy
 - ☐ E) Nerve conduction studies



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11. A 47-year-old woman is brought to the physician because of progressive loss of visual acuity over the past 6 months. Snellen chart testing shows a visual acuity of 20/180. A photograph of the optic fundus is shown. The remainder of the examination shows no abnormalities. Which of the following is the most likely cause of these findings?

- ☐ A) Acute glaucoma
- ☐ B) Detached retina
- ☐ C) Diabetic retinopathy
- ☐ D) Macular degeneration
- ☐ E) Optic neuritis



12. A 42-year-old man, who immigrated to the USA 3 years ago, tells his physician that his hands are numb. He is being treated with isoniazid for tuberculosis. His features are mildly coarsened. Examination shows widespread patches of loss of pinprick and light touch sensations, especially over the tip of the nose and the ears. The patches are oval or irregular and measure 2 to 3 cm. The common peroneal nerves feel thickened. Which of the following is the most likely diagnosis?
- ☐ A) Acute intermittent porphyria
 - ☐ B) Alcoholic neuropathy
 - ☐ C) Charcot-Marie-Tooth disease
 - ☐ D) Diabetic amyotrophy
 - ☐ E) Guillain-Barré syndrome
 - ☐ F) Leprosy
 - ☐ G) Motor neuropathy with conduction block
 - ☐ H) Paraneoplastic neuropathy
 - ☐ I) Paraproteinemic neuropathy



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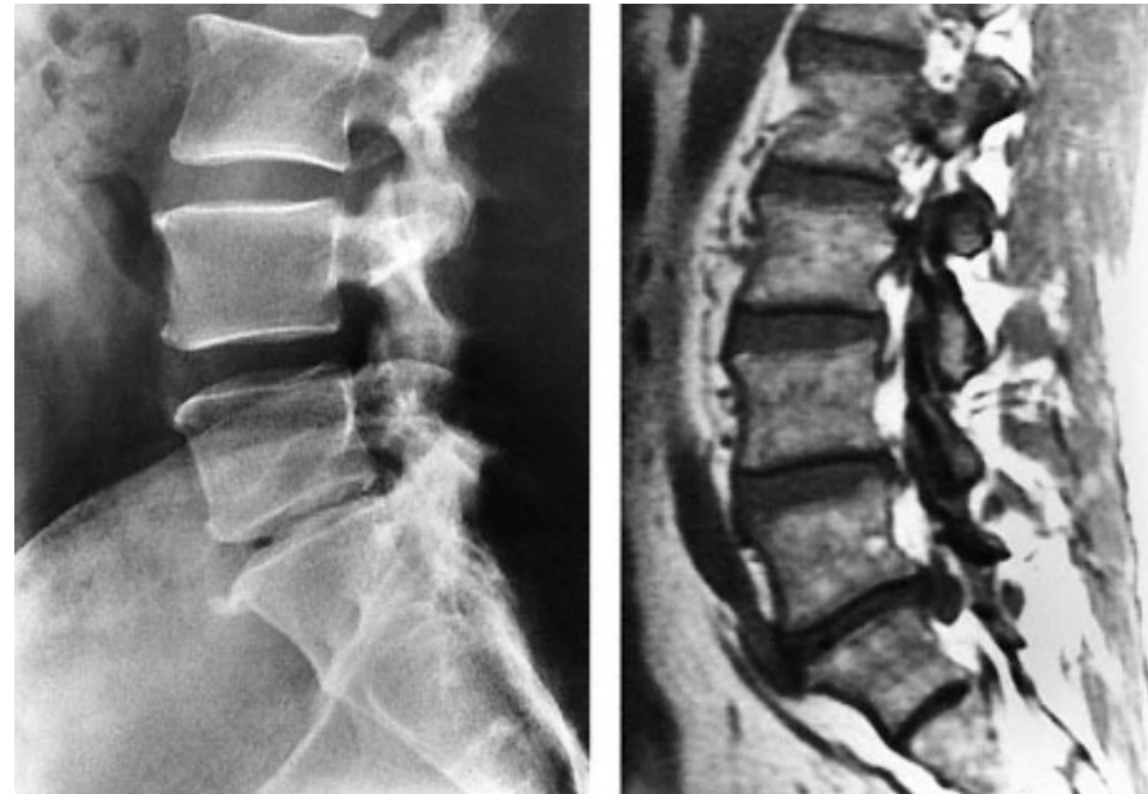
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13. A 77-year-old man is brought to the emergency department because of confusion, lethargy, and excessive sleepiness over the past 3 weeks. Three days ago, he fell and struck his head. On arrival, he has a generalized tonic-clonic seizure that lasts 40 seconds. He has hypertension and type 2 diabetes mellitus and has a history of frequent urinary tract infections and renal calculi. His medications include losartan, glyburide, and trimethoprim-sulfamethoxazole. He is lethargic but easily aroused. His temperature is 37°C (98.6°F), pulse is 110/min, respirations are 20/min, and blood pressure is 190/105 mm Hg. Examination shows an ecchymosis over the left scalp. Pupils are equally round and reactive to light. Doll's eye (oculocephalic) maneuver shows full eye movement. Cranial nerves are intact. He moves all four extremities in response to painful stimuli. Deep tendon reflexes are absent in the upper and lower extremities. Babinski sign is absent bilaterally. There is an occasional myoclonic jerk in the lower extremities. He is disoriented to place and time. Laboratory studies show:

Hemoglobin	9 g/dL
Leukocyte count	7500/mm ³
Serum	
Urea nitrogen	105 mg/dL
Creatinine	5.3 mg/dL
Alkaline phosphatase	150 U/L
ALT	47 U/L

Which of the following is the most likely cause of the patient's lethargy?

- ☐ A) Hepatic failure
- ☐ B) Pontine hematoma
- ☐ C) Status epilepticus
- ☐ D) Subdural hematoma
- ☐ E) Uremic encephalopathy



14. For 10 days, a 40-year-old man has had pain in his left buttock and posterior thigh that is exacerbated by coughing or sneezing. Deep tendon reflexes are absent at the left ankle. Passive flexion of the right hip with the knee extended produces pain in the left lower extremity. An x-ray and an MRI are shown. Which of the following is the most likely diagnosis?

- ☐ A) Fracture of the pars interarticularis
- ☐ B) Herniated nucleus pulposus
- ☐ C) Mechanical low back strain
- ☐ D) Metastatic carcinoma
- ☐ E) Spinal stenosis

15. A 52-year-old man comes to the physician because of a 3-month history of insomnia. He has had difficulty falling asleep because of restlessness and a feeling of "ants crawling inside my legs." He notes that he has to constantly move his legs to relieve the sensation. He often has to uncover his legs or hang them over the side of the bed to get some relief. These symptoms now occur almost nightly. He has milder symptoms during the day but is generally moving around so the sensation is not as noticeable. Physical examination shows no abnormalities. Laboratory studies show:

Hemoglobin	13.5 g/dL
Hemoglobin A _{1c}	5.6%
Serum ferritin	15 ng/mL

Thyroid function tests show no abnormalities. Which of the following is the most appropriate pharmacotherapy?

- ☐ A) Imipramine
- ☐ B) Paroxetine
- ☐ C) Ropinirole
- ☐ D) Trazodone
- ☐ E) Triazolam



16. A 57-year-old man comes to the physician because of increasing problems with memory. He reports that he was fired from his job as an accountant recently because he was no longer able to maintain accurate accounts. He has a 5-year history of facial grimacing and involuntary arm and leg spasms that have become increasingly severe. He has hypertension treated with captopril. There is no family history of neurologic illness. The patient's father died at the age of 28 years of injuries sustained in a motor vehicle collision. The patient has never smoked cigarettes, and he does not drink alcohol or use illicit drugs. His temperature is 36.8°C (98.2°F), pulse is 80/min, respirations are 20/min, and blood pressure is 142/82 mm Hg. Neurologic examination shows frequent, asymmetric, involuntary movements of all extremities that are random and jerk-like. On mental status examination, the patient recalls zero of three objects after 5 minutes; he cannot name the current president or recall the meaning of proverbs. An MRI of the brain is most likely to show which of the following?

- ☐ A) Atrophy of the caudate nucleus
- ☐ B) Atrophy of the substantia nigra
- ☐ C) Atrophy of the thalamus
- ☐ D) Increased T2 signal in the globus pallidus
- ☐ E) Increased T2 signal in the putamen
- ☐ F) Increased T2 signal in the subthalamic nuclei



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17. A 47-year-old woman is brought to the emergency department 1 hour after the sudden onset of headache. She says that she was walking from work to her car when she felt an explosion in her head causing her to fall to the ground. She is not sure if she lost consciousness. Since then, she has had a severe headache characterized by double vision and throbbing pain exacerbated by head movement. She has hypertension treated with atenolol. She has smoked two packs of cigarettes daily for 20 years. Her pulse is 80/min, respirations are 16/min, and blood pressure is 148/82 mm Hg. Examination shows left ptosis. The left pupil is 6.5 mm, and the right pupil is 4.5 mm. Upward gaze and adduction of the left eye are limited. A CT scan of the head shows diffuse subarachnoid blood. A lumbar puncture yields grossly bloody cerebrospinal fluid. Cerebral angiography is most likely to show a saccular aneurysm in which of the following locations?

- ☐ A) Basilar artery
- ☐ B) Bifurcation of the left middle cerebral artery
- ☐ C) Left posterior communicating artery
- ☐ D) Origin of left anterior cerebral artery
- ☐ E) Origin of left ophthalmic artery



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18. A 72-year-old woman comes to the physician because of a 4-week history of involuntary movements of her tongue and jaw. She has a 3-year history of intermittent nausea. Medications include psyllium, metoclopramide, ranitidine, calcium carbonate, and bismuth subsalicylate. Examination shows licking movements of the tongue and side-to-side movements of the jaw. There is mild slowness of limb movements. Which of the following drugs is the most likely cause of these findings?

- ☐ A) Bismuth subsalicylate
- ☐ B) Calcium carbonate
- ☐ C) Metoclopramide
- ☐ D) Psyllium
- ☐ E) Ranitidine



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19. A 22-year-old woman, gravida 2, para 1, aborta 1, comes to the physician because of two episodes of blurred vision in the left eye and one episode of right-sided weakness during the past 3 years. She also has had urinary urgency and frequency with some episodes of incontinence during this time. The first episode of blurred vision occurred after a motor vehicle collision 3 years ago during which she sustained a fracture of her left humerus and soft-tissue contusions. She has used an oral contraceptive for the past 2 years. She has smoked one pack of cigarettes daily for 3 years. She is of northern European descent and has lived in the southern United States for her entire life. Vital signs are within normal limits. Funduscopic examination shows mild pallor of the left optic disc. Visual acuity is 20/80 on the left and 20/25 on the right. Neurologic examination shows mild right hemiparesis and bilateral ataxia on finger-nose and heel-shin testing. The remainder of the examination shows no abnormalities. An MRI of the brain shows multiple areas of increased T2 signal in the periventricular white matter and the white matter of the cerebellum. Which of the following is the strongest predisposing factor for this patient's condition?

- ☐ A) Ancestry
- ☐ B) History of spontaneous abortion
- ☐ C) History of trauma
- ☐ D) Residence in warm climate
- ☐ E) Use of oral contraceptives



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20. Four days after hospitalization for internal fixation of a femoral fracture sustained in a motor vehicle collision, a 32-year-old man becomes agitated and disruptive. His temperature is 38°C (100.4°F), pulse is 124/min, respirations are 24/min, and blood pressure is 180/100 mm Hg. He is diaphoretic and picking at his bedsheets. Funduscopic and motor and sensory examinations show no abnormalities. Mental status examination shows psychomotor agitation and disorientation. His hematocrit is 40%, and leukocyte count is 7000/mm³. Which of the following is the most likely diagnosis?

- ☐ A) Cryptococcal meningitis
- ☐ B) Delirium tremens
- ☐ C) Fat embolism
- ☐ D) Hypoglycemia
- ☐ E) Subdural hematoma



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21. A 57-year-old woman with metastatic breast cancer comes to the physician for a follow-up examination. She reports that her pain is poorly controlled with her current dosage of oral sustained-release morphine. One month ago, she reported the same problem and the dosage of morphine was increased. On examination today, she is agitated and tearful and insists on another increase in her morphine dosage. Which of the following is the most likely explanation for this patient's request?

- ☐ A) Inadequate pain control
- ☐ B) Intent to attempt suicide
- ☐ C) Major depressive disorder
- ☐ D) Substance abuse
- ☐ E) Surreptitious use of morphine by family



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22. A 62-year-old man comes to the physician because of a 9-month history of weakness in his left arm and light-headedness when he plays tennis. His left hand and arm are cool to the touch during these episodes, and his symptoms usually resolve after 15 minutes of rest. He plays tennis for 60 minutes three times weekly, and the episodes occur twice weekly. He is left-handed. He does not recall any trauma to his left arm and has not had any other symptoms. He smoked one and one-half packs of cigarettes daily for 27 years but stopped 10 years ago. He drinks one glass of wine daily. His pulse is 78/min and regular, and blood pressure is 150/88 mm Hg in the right arm and 136/70 mm Hg in the left arm. The lungs are clear to auscultation. S_1 and S_2 are normal. The left radial pulse is decreased. The remainder of the examination, including neurologic examination, shows no abnormalities. Which of the following is the most appropriate next step in diagnosis?

- ☐ A) Ambulatory ECG monitoring
- ☐ B) Tilt test
- ☐ C) Echocardiography
- ☐ D) Cardiac stress scintigraphy
- ☐ E) MR angiography



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23. A 35-year-old man with a 3-year history of dermatomyositis has had progressive weakness for 3 weeks. He had previously responded well to high-dose prednisone therapy. He appears cushingoid; serum creatine kinase and aldolase activities are now normal. Which of the following is the most likely diagnosis?

- ☐ A) Corticosteroid myopathy
- ☐ B) Hypothyroidism
- ☐ C) Major depressive disorder
- ☐ D) Peripheral neuropathy
- ☐ E) Progressive myositis



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24. A 72-year-old woman is brought to the physician because of increasing clumsiness, restlessness, and confusion over the past 3 weeks. She currently lives with her daughter. She can no longer remember how to make coffee or button her clothes. She has urinary incontinence. Her medications include digoxin, furosemide, potassium, ibuprofen, ranitidine, and diphenhydramine. One month ago, she began taking meperidine for a rotator cuff injury. Examination shows disorientation, memory loss, and dyspraxia. A CT scan of the head shows mild cortical atrophy. Which of the following is the most appropriate next step in management?

- ☐ A) Discontinue anticholinergic medications
- ☐ B) Discuss obtaining durable power of attorney
- ☐ C) EEG
- ☐ D) Fluorescent treponemal antibody assay
- ☐ E) Prescribe an antidepressant
- ☐ F) Recommend 24-hour in-home nursing assistance
- ☐ G) Suggest labeling the drawers and cabinets for contents



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25. A 62-year-old man is brought to the physician by his wife because of progressive behavior changes over the past 8 weeks. His initial symptoms included anxiety and fatigue that were followed by decreased memory and disorientation. Two weeks later, he developed progressive imbalance that has progressed to the point where he is unable to walk unassisted. Over the past week, he has had brief body jerks that are precipitated by loud noises. Examination shows disorientation, decreased memory, inability to walk due to marked ataxia, and stimulus-induced myoclonus. An MRI of the brain shows generalized cerebral atrophy. Which of the following is the most likely diagnosis?

- ☐ A) Benzodiazepine intoxication
- ☐ B) Creutzfeldt-Jakob disease
- ☐ C) Dementia, Alzheimer type
- ☐ D) Hydrocephalus
- ☐ E) Hypothyroidism
- ☐ F) Multi-infarct (vascular) dementia
- ☐ G) Progressive supranuclear palsy
- ☐ H) Wernicke-Korsakoff syndrome



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26. A 72-year-old man is brought to the emergency department because of a decreased level of consciousness for the past 6 hours. Three days ago, he had fever, shortness of breath, and productive cough treated with an antibiotic, but his symptoms did not improve. On arrival, his temperature is 39°C (102.2°F), pulse is 110/min, respirations are 28/min, and blood pressure is 110/75 mm Hg. Breath sounds are decreased over the right midlung field. On neurologic examination, he is unarousable but responds to tactile stimuli by moaning. Cranial nerves are intact. There is resistance to passive flexion of the neck. Which of the following is the most likely pathogen?

- ☐ A) Herpes simplex virus 1
- ☐ B) *Listeria monocytogenes*
- ☐ C) *Pseudomonas aeruginosa*
- ☐ D) *Streptococcus pneumoniae*
- ☐ E) *Toxoplasma gondii*



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27. A 37-year-old woman comes to the physician because she has stumbled several times during the past 4 days. She has also noticed intermittent numbness of her left shin and foot. She has no history of serious illness and takes no medications. She delivered her two children after uncomplicated pregnancies 8 and 10 years ago. Vital signs are within normal limits. Muscle strength is 3/5 on dorsiflexion of the left foot. Range of motion of the hip does not produce any symptoms. Deep tendon reflexes are normal. Sensation to pinprick is decreased over the lateral aspect of the left shin and the dorsal surface of the left foot. The remainder of the examination shows no abnormalities. Which of the following is the most appropriate initial recommendation?

- ☐ A) Avoiding sitting with crossed legs
- ☐ B) Starting an exercise program
- ☐ C) Wearing flat shoes that fit comfortably
- ☐ D) Placing an orthotic pad in the left shoe
- ☐ E) Taking nonsteroidal anti-inflammatory drugs



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28. An 82-year-old woman with osteoarthritis is brought to the emergency department because of a 4-day history of increasing confusion and progressive generalized weakness. During the past 2 months, she has fallen several times; there was no loss of consciousness. She has had no difficulty sleeping. Two weeks ago, hydrochlorothiazide and metoprolol were started for hypertension. Her temperature is 36.7°C (98°F), pulse is 68/min, and blood pressure is 140/86 mm Hg. Examination shows a right pronator drift. Babinski sign is present on the right. She is drowsy, inattentive to questions, and gives several inaccurate answers. Which of the following is the most likely explanation for this patient's mental status changes?

- ☐ A) Adverse effect of metoprolol
- ☐ B) Cerebral infarction
- ☐ C) Herpes encephalitis
- ☐ D) Hypothyroidism
- ☐ E) Subdural hematoma



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29. A previously healthy 58-year-old man is admitted to the hospital because of three episodes of amaurosis fugax in the right eye and two episodes of temporary weakness of the left hand and arm over the past 3 weeks. Which of the following is the most likely cause?

- ☐ A) Occlusion of the right common carotid artery
- ☐ B) Occlusion of the left internal carotid artery
- ☐ C) Occlusion of the right vertebral artery
- ☐ D) Ulcerative plaque of the basilar artery
- ☐ E) Ulcerative plaque of the right internal carotid artery



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30. A previously healthy 18-year-old man is brought to the emergency department because of facial weakness and ear pain that began this morning. When he awoke this morning, he looked in the mirror and noticed that his face appeared twisted. While he was eating breakfast, milk drooled from the left side of his mouth. The ear pain has not been relieved by acetaminophen. Examination shows weakness of both the upper and lower face on the left. Sensation over the face is normal. Which of the following is the most likely location of the abnormality?

- ☐ A) Anterior horn cell
- ☐ B) Muscle
- ☐ C) Myoneural junction
- ☐ D) Nerve
- ☐ E) Nerve root



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31. A 22-year-old college student has a 4-year history of severe, throbbing, generalized headaches that occur once weekly. The headaches are preceded by a 20-minute period of seeing bright spots. Which of the following drugs is most likely to be effective in reducing the frequency of this patient's headaches?

- ☐ A) Anticholinergic drugs
- ☐ B) β -Adrenergic blocking drugs
- ☐ C) Ergotamine tartrate
- ☐ D) Salicylates
- ☐ E) Sympathomimetic drugs



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32. A 62-year-old woman comes to the physician because of right-sided facial pain for 3 days and a facial rash for 1 day. She describes the pain as a burning uncomfortable feeling in her cheek. She had cold symptoms 1 week ago that have now resolved. She has hypertension treated with hydrochlorothiazide but is otherwise healthy. She cares for her two grandchildren, ages 1 and 3 years, every weekday after school. Examination shows a vesicular rash over the face. Neurologic examination shows no abnormalities. Which of the following is the most appropriate measure to prevent transmission to the unvaccinated grandchildren?

- ☐ A) Avoid contact with the grandmother until the rash has resolved
- ☐ B) Administration of varicella vaccine to the children
- ☐ C) Acyclovir therapy for the children
- ☐ D) Acyclovir therapy for the grandmother
- ☐ E) No preventive measures available



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33. A 17-year-old girl is brought to the emergency department 30 minutes after striking her forehead on the dashboard in a motor vehicle collision. Examination shows lethargy and confusion. A CT scan of the head shows a frontal lobe injury. Which of the following is the most likely manifestation of this injury?

- ☐ A) Dysdiadochokinesia
- ☐ B) Ideomotor apraxia
- ☐ C) Problems with conceptual planning
- ☐ D) Prosopagnosia
- ☐ E) Spatial disorientation



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34. A 57-year-old woman is brought to the physician because of a 4-week history of progressive tingling in her hands and feet. She says she has also had decreased sensation in her feet, "like I'm walking on sponges," and a painless tingling sensation down her back and legs when she flexes her neck. She has not had weakness or changes in bowel and bladder function. She was diagnosed with stage III ovarian cancer 6 months ago and has been treated with intravenous cisplatin and etoposide monthly. She has hypothyroidism treated with levothyroxine. Muscle strength is normal. Deep tendon reflexes are absent. Babinski sign is absent. Sensation to vibration is decreased in the fingertips and absent below the knees, and sensation to pinprick is decreased in a stocking distribution. Proprioception is decreased in the fingers and toes. Romberg sign is present. Her gait is slightly wide based. Coordination is normal. Which of the following is the most likely cause of this patient's neurologic symptoms?

- ☐ A) Guillain-Barré syndrome
- ☐ B) Hypothyroidism
- ☐ C) Meningeal carcinoma
- ☐ D) Multiple sclerosis
- ☐ E) Toxic neuropathy



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35. A 6-month-old girl is brought to the physician for a follow-up examination after two audiometry screenings showed hearing loss. She was born at term to a 27-year-old primigravid woman after an uncomplicated pregnancy. Examination of the infant today shows no abnormalities. Audiometry today shows significant hearing loss bilaterally. Which of the following is the most likely cause of this infant's condition?

- ☐ A) Cytomegalovirus infection
- ☐ B) Group B β -streptococcal infection
- ☐ C) Herpes simplex
- ☐ D) Parvovirus infection
- ☐ E) Respiratory syncytial virus



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36. A 57-year-old man is brought to the physician by his wife because of constant fatigue for 2 weeks. He has consumed 1 pint of vodka daily for 30 years. He appears drowsy. His temperature is 37°C (98.6°F), pulse is 100/min, respirations are 20/min, and blood pressure is 120/76 mm Hg. Cardiopulmonary examination shows no abnormalities. Abdominal examination shows severe distention with a fluid wave. Rectal examination shows hemorrhoids. Test of the stool for occult blood is positive. On mental status examination, he responds to questions slowly. When asked to write his name, it is an unintelligible scribble compared with a clear signature on his medical chart from 1 year ago. Laboratory studies show:

Hemoglobin	12.5 g/dL
Hematocrit	39%
Leukocyte count	7000/mm ³ with a normal differential
Platelet count	200,000/mm ³
Prothrombin time	18 sec (INR=2)
Partial thromboplastin time	40 sec
Serum	
Urea nitrogen	6 mg/dL
Creatinine	0.6 mg/dL
AST	8 U/L
ALT	8 U/L

Which of the following is most likely to improve this patient's mental status?

- ☐ A) High-protein, low-carbohydrate diet
- ☐ B) Administration of lactulose
- ☐ C) Infusion of fresh frozen plasma
- ☐ D) Transfusion of packed red blood cells
- ☐ E) Hemodialysis
- ☐ F) Paracentesis



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37. A 32-year-old woman comes to the physician because of weakness of her right arm and leg that began 1 week ago when she awoke in the morning. Six days ago, when she took a bath, she noticed that she was unable to feel the heat of the water on the left side of her body. She had a mild upper respiratory tract infection 3 weeks ago. Cranial nerves are intact. There is mild weakness of the right upper and lower extremities. Sensation to pinprick and temperature is decreased over the left trunk and left upper and lower extremities; sensation to vibration is decreased over the right foot. Reflexes in the upper and lower extremities are more brisk on the right than the left. Babinski sign is present on the right. Which of the following is the most likely location of this patient's lesion?

- ☐ A) Brain stem
- ☐ B) Cauda equina
- ☐ C) Cerebral hemisphere
- ☐ D) Cervical nerve root
- ☐ E) Lumbar nerve root
- ☐ F) Peripheral nerve
- ☐ G) Spinal cord



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38. A 72-year-old man with a 3-year history of Parkinson disease is brought to the physician by his wife for a follow-up examination. Three weeks ago, his dosage of carbidopa-levodopa was increased. Since then, he has reported seeing people spying on him from across the street. He has no other history of serious medical or psychiatric illness and takes no other medications. Physical examination shows a resting tremor. There is cogwheel rigidity and increased muscle tone. On mental status examination, he reports seeing people spying on him and says his wife is "one of them." Addition of which of the following medications is the most appropriate next step in pharmacotherapy?

- ☐ A) Haloperidol
- ☐ B) Lorazepam
- ☐ C) Paroxetine
- ☐ D) Quetiapine
- ☐ E) Valproic acid



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39. A 72-year-old woman is brought to the physician by her son because of a 1-month history of progressive forgetfulness. During this period, she has misplaced her keys and checkbook, and she forgot that she was supposed to meet him for lunch last week. The patient says that she has had progressive fatigue because of recent difficulty staying asleep at night. She often awakens at 4 or 5 am and is unable to fall back asleep. She takes zolpidem once or twice weekly to help her sleep, but she has taken it nightly for the past 2 weeks. She also has had a 4.5-kg (10-lb) weight loss and says that nothing tastes good anymore. She has a 10-year history of esophageal reflux disease treated with lansoprazole and a 2-year history of arthritis treated with acetaminophen. She was an English professor at a prominent university until 1 year ago when she was required to retire by university regulations. She maintained an active schedule of lecturing and writing until 2 months ago. Her hair is unwashed, and she has a mild body odor. She is 168 cm (5 ft 6 in) tall and weighs 50 kg (110 lb); BMI is 18 kg/m². Her pulse is 72/min, and blood pressure is 125/70 mm Hg. On mental status examination, she appears distracted and worried. She says that her mood is "okay." Her speech is slow and monotone, and she answers with single words. She is alert and oriented to person, place, and time. Her digit span is 6 forward and 2 backward. She can recall zero of three objects after 5 minutes. She refuses to perform simple change calculations or name past presidents. Which of the following is the most likely diagnosis?

- ☐ A) Adjustment disorder with depressed mood
- ☐ B) Delirium
- ☐ C) Dementia, Alzheimer type
- ☐ D) Dysthymic disorder
- ☐ E) Major depressive disorder
- ☐ F) Parkinson disease
- ☐ G) Substance-induced persisting amnesic disorder



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40. A 77-year-old man with Parkinson disease is brought to the physician by his wife because of a 3-month history of unusual behavior at night. She says that he yells in his sleep and waves his arms as if defending himself. Two nights ago, he bit her and tried to push her out of the bed. Current medications include carbidopa-levodopa. Examination shows little facial expression, a pill-rolling tremor of both hands, and muscle rigidity. Which of the following is the most appropriate next step in diagnosis?

- ☐ A) Multiple sleep latency testing
- ☐ B) Overnight pulse oximetry testing
- ☐ C) Polysomnography
- ☐ D) Sleeping and waking EEG
- ☐ E) Wrist actigraphy



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41. A 42-year-old woman is brought to the emergency department by her girlfriend 15 minutes after several episodes of staring and blinking while eating dinner. The patient has a history of bipolar disorder well controlled with daily bupropion and lithium carbonate for the past 5 years. She started taking terbinafine 2 weeks ago for chronic toenail fungus. During the examination, she is conversant. Her pulse is 64/min, respirations are 16/min, and blood pressure is 140/90 mm Hg. A normal rhythm is heard on cardiac examination. Neurologic examination shows no focal findings. Which of the following is the most likely explanation for this patient's symptoms?

- ☐ A) Catatonic episode
- ☐ B) Glioblastoma multiforme
- ☐ C) Rapid cycling bipolar disorder
- ☐ D) Seizure activity
- ☐ E) Transient ischemic attack



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42. A 27-year-old man who is HIV positive comes to the physician because of left arm clumsiness for 1 week. His CD4+ T-lymphocyte count was $78/\text{mm}^3$ (Normal ≥ 500) 8 weeks ago. Medications include zidovudine (AZT), didanosine, ritonavir, and azithromycin. He appears well. His temperature is 37.2°C (99°F), pulse is 96/min, and respirations are 16/min. Examination shows weakness of the left upper extremity. A CT scan of the head with contrast shows a ring-enhancing lesion in the right parietal cortex. Which of the following is the most appropriate next step in management?

- ☐ A) Intravenous methylprednisolone therapy
- ☐ B) Oral dapsone therapy
- ☐ C) Oral sulfadiazine-pyrimethamine therapy
- ☐ D) Oral trimethoprim-sulfamethoxazole therapy
- ☐ E) Lumbar puncture
- ☐ F) Brain biopsy



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43. A previously healthy 32-year-old woman had the acute onset of severe headache followed by somnolence 2 hours ago. Her temperature is 38.3°C (101°F), pulse is 96/min, and blood pressure is 140/92 mm Hg. She is lethargic but oriented and answers questions appropriately. A CT scan of the head shows no abnormalities except for small ventricles. Laboratory studies show:

Serum glucose	90 mg/dL
Cerebrospinal fluid	
Color	pink
Opening pressure	25 mm H ₂ O
Glucose	53 mg/dL
Protein	85 mg/dL
RBC	4000/mm ³
WBC	10/mm ³
Segmented neutrophils	65%
Lymphocytes	35%

Which of the following is the most likely diagnosis?

- ☐ A) Cryptococcal meningitis
- ☐ B) Enterovirus infection
- ☐ C) Idiopathic intracranial hypertension
- ☐ D) Subarachnoid hemorrhage
- ☐ E) West Nile encephalitis

44. A 62-year-old woman with metastatic breast cancer is brought to the emergency department because of urinary incontinence for 24 hours. She has metastases to the skull, multiple vertebral bodies, ribs, liver, and para-aortic lymph nodes. Medications include tamoxifen, sustained-release morphine, stool softeners, and fluoxetine. Her temperature is 36.7°C (98.1°F), pulse is 84/min, respirations are 14/min, and blood pressure is 128/76 mm Hg. Examination shows a large suprapubic mass. There is significant pain over the entire lumbar spine on fist percussion. Rectal examination shows decreased sphincter tone. Which of the following is the most likely cause of the incontinence?

- ☐ A) Anticholinergic effects of fluoxetine
- ☐ B) Hypercalcemia
- ☐ C) Metastases to the bladder
- ☐ D) Metastases to the cauda equina
- ☐ E) Urethral spasm



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45. A 52-year-old woman is brought to the emergency department by her husband 30 minutes after a generalized tonic-clonic seizure. Her husband reports that she has had a headache and has seemed confused over the past 3 days. She was scheduled to be evaluated by a pulmonologist later this week for the recent onset of mild hemoptysis. She smoked two packs of cigarettes daily for 30 years but quit 5 years ago. She has a 5-year history of type 2 diabetes mellitus controlled by diet. On arrival, she is postictal. Her temperature is 37°C (98.6°F), and blood pressure is 110/60 mm Hg. She is minimally responsive to painful stimuli. Cardiopulmonary examination shows no abnormalities. There is no peripheral edema. There are no focal neurologic deficits. Laboratory studies show:

Serum	
Na ⁺	112 mEq/L
Glucose	150 mg/dL
Creatinine	0.7 mg/dL
Cholesterol	230 mg/dL
Osmolality	238 mOsmol/kg
Urine	
Glucose	trace
Protein	none
RBC	1–3/hpf
WBC	1–3/hpf
Osmolality	240 mOsmol/kg
Na ⁺	25 mEq/L

Which of the following is the most likely diagnosis?

- | | |
|---|---|
| ☐ A) Bacterial pneumonia | ☐ H) Intracranial neoplasm |
| ☐ B) Cirrhosis of the liver | ☐ I) Meningitis |
| ☐ C) Decreased solute intake | ☐ J) Nephrotic syndrome |
| ☐ D) Dehydration | ☐ K) Pseudohyponatremia |
| ☐ E) Glucocorticoid deficiency | ☐ L) Psychogenic polydipsia |
| ☐ F) Hyperglycemia | ☐ M) Renal failure |
| ☐ G) Inappropriate secretion of ADH (vasopressin) | |



46. A 3-week-old male newborn is brought to the emergency department by his parents because of a 1-day history of increased fussiness and drowsiness. En route to the emergency department, he had rhythmic jerking movements of all extremities. His mother received minimal prenatal care and was treated for gonorrheal cervicitis during pregnancy; her *Streptococcus agalactiae* (group B) status was unknown. He was born at 38 weeks' gestation. Examination at birth showed no abnormalities. Today, he is irritable and responds only to painful stimulation. His temperature is 39°C (102.2°F), pulse is 160/min, and respirations are 32/min. Examination shows no conjunctivitis or cutaneous lesions. The lungs are clear to auscultation. A lumbar puncture is performed, and cerebrospinal fluid analysis shows:

Glucose	60 mg/dL
Total protein	100 mg/dL
WBC	300/mm ³
RBC	5000/mm ³
Gram stain	
WBC	2+
Organisms	none

In addition to ceftriaxone and vancomycin, which of the following is the most appropriate pharmacotherapy?

- ☐ A) Acyclovir
- ☐ B) Dexamethasone
- ☐ C) Ganciclovir
- ☐ D) Ribavirin
- ☐ E) Zidovudine



47. An otherwise healthy 62-year-old man is brought to the emergency department 45 minutes after an episode of loss of consciousness while shaving. Prior to the episode, he had light-headedness, nausea, and dimming of vision. His wife states that he was unconscious for about 1 minute and lost control of bladder function. She did not notice any movement during the episode. He had a similar episode 1 month ago while shaving. At that time, echocardiography, 24-hour ambulatory ECG monitoring, and a tilt test showed no abnormalities; cardiac enzyme activities were within the reference range. His current blood pressure is 150/96 mm Hg. A grade 2/6, systolic ejection murmur is heard best at the upper left sternal border. The remainder of the examination shows no abnormalities. Which of the following is the most likely diagnosis?

- ☐ A) Absence seizure
- ☐ B) Aortic stenosis
- ☐ C) Carotid sinus hypersensitivity
- ☐ D) Generalized tonic-clonic seizure
- ☐ E) Neurocardiogenic (vasovagal) syncope
- ☐ F) Transient ischemic attack



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48. A 32-year-old man is brought to the physician because of a 10-month history of difficulty walking and decreased grip strength and stiffness of his hands that are interfering with his job as a plumber. He has no history of serious illness. There is frontal balding. Cranial nerve examination shows bilateral ptosis and atrophy of the sternocleidomastoid and temporalis muscles. There is distal intrinsic musculature weakness of the hand and bilateral weakness on ankle dorsiflexion. Percussion of the hypothenar muscles results in slow relaxation. Sensation is intact. Which of the following is the most likely diagnosis?
- ☐ A) Amyotrophic lateral sclerosis
 - ☐ B) Facioscapulohumeral muscular dystrophy
 - ☐ C) Myasthenia gravis
 - ☐ D) Myotonic dystrophy
 - ☐ E) Spinal muscular atrophy



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49. A 57-year-old woman comes to the physician because of persistent numbness of the back of her right hand for 2 weeks. Her symptoms began after operative treatment of a distal radial fracture. Sensation to pinprick is decreased over the dorsal first web, thumb, and index finger. There is no motor deficit. Which of the following is the most likely site of nerve injury?

- | | |
|---|---|
| ☐ A) Axillary nerve | ☐ I) Radial nerve above the elbow |
| ☐ B) Long thoracic nerve | ☐ J) Radial nerve at the elbow |
| ☐ C) Median nerve above the elbow | ☐ K) Radial nerve at the distal forearm |
| ☐ D) Median nerve at the elbow | ☐ L) Suprascapular nerve |
| ☐ E) Median nerve at the wrist | ☐ M) Thoracodorsal nerve |
| ☐ F) Musculocutaneous nerve above the elbow | ☐ N) Ulnar nerve above the elbow |
| ☐ G) Musculocutaneous nerve at the elbow | ☐ O) Ulnar nerve at the elbow |
| ☐ H) Musculocutaneous nerve at the wrist | ☐ P) Ulnar nerve at the wrist |



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50. A 62-year-old woman is brought to the emergency department because of a 3-day history of progressive weakness and numbness. These symptoms first occurred in her feet and have spread to include her chest. She also has had difficulty emptying her bladder completely. She has moderate midback pain that began 1 month ago and now radiates to her chest; it is exacerbated when she coughs. One year ago, she had breast cancer and underwent lumpectomy followed by radiation and chemotherapy. Her only medication is tamoxifen. Vital signs are within normal limits. Muscle strength is 5/5 in the upper extremities and 4/5 in the lower extremities with increased tone. Deep tendon reflexes are 2+ in the upper extremities and 4+ in the lower extremities. Babinski sign is present bilaterally. Sensation to pinprick is decreased below the T4 level. Which of the following is the most likely diagnosis?

- ☐ A) Epidural spinal cord compression
- ☐ B) Guillain-Barré syndrome
- ☐ C) Leptomeningeal carcinomatosis
- ☐ D) Paraneoplastic myelopathy
- ☐ E) Transverse myelitis



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